XRPL Validator — Daily Engineering Report · 2026-09-09

Period: 2026-09-09 00:00 to 24:00 US Eastern (2026-09-09 04:00 to 2026-09-10 04:00 UTC); ledgers #106,859,725 to #106,882,087. Every count from the ledger and the validator's logs stops at #106,882,087. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 14 engine receipts recorded (8 fixed, 0 explained, 6 without a recorded fix).
- **Changes:** 0 deployments, 13 commits merged into main, 14 engine findings fixed (F229–F242).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,859,725–#106,882,087), by step:

Step	Ledgers with a receipt
Step 1 · native engine	14
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,860,967	Step 1 · native engine	1 object(s) differed in content	fixed: F238, commit 1274170, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-02	—	#106,863,376	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: CA462483 (deleted), ACA888DC, AD3A9337	fixed: F239, commit 0ed3d96, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-03	—	#106,864,621	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-04	—	#106,867,574	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: 370BE1EA; 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F240, commit 5d1a89d, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-05	—	#106,868,992	Step 1 · native engine	28 transaction result(s) differed from rippled's	fixed: F241, commit e37e205, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-06	—	#106,869,040	Step 1 · native engine	28 transaction result(s) differed from rippled's	no disposition recorded
R-07	—	#106,869,060	Step 1 · native engine	28 transaction result(s) differed from rippled's	no disposition recorded
R-08	—	#106,869,297	Step 1 · native engine	1 object(s) differed in content	fixed: F242, commit 3fefb0d, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-09	—	#106,869,323	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-10	—	#106,871,187	Step 1 · native engine	1 object(s) differed in content	fixed: F243, commit 6328e1b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-11	—	#106,873,753	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F244, commit 60b4142, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-12	—	#106,874,381	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F244, commit 60b4142, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-13	—	#106,874,385	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-14	—	#106,874,386	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-09-01, INC-01 is INC-2026-09-09-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
11:58 UTC	xrpl-validator	`1274170`	fix(xrpl-ledger): F238 — an owner-directory insert sorts the whole page, not just the newcomer's slot
11:58 UTC	xrpl-validator	`0ed3d96`	fix(xrpl-ledger): F239 — a bridged strand's rev pass reaps past the level it consumes whole
11:58 UTC	xrpl-validator	`ab1a5c4`	fix(xrpl-ledger): F237 — a bridged round takes every leg-A offer at the tip's quality into one leg-B slice
11:58 UTC	xrpl-validator	`1ad42e1`	fix(xrpl-ledger): F236 — the direct walk rests an offer's in side as TakerGets less the gross fold divided once by the gateway rate
11:58 UTC	xrpl-validator	`c6751ea`	fix(xrpl-ledger): F235 — NFTokenCreateOffer's preclaim is rippled's tokenOfferCreatePreclaim: an unfunded buyer is tecUNFUNDED_OFFER
11:58 UTC	xrpl-validator	`d959488`	fix(xrpl-ledger): F234 — CheckCash cashes through the payment engine: SendMax buys what it can, DeliverMin is a floor, XRP shortfall is tecUNFUNDED_PAYMENT
11:58 UTC	xrpl-validator	`80eaadd`	fix(xrpl-ledger): F233 — bridge peeks reap dead offers only on the strands rippled flows this round
11:58 UTC	xrpl-validator	`f7ef2ee`	fix(xrpl-ledger): F232 — a pool turn that ends the iteration leaves the tip in place; judge the beyond-strict sweep there, post-turn
11:58 UTC	xrpl-validator	`e10f8d2`	fix(xrpl-ledger): F231 — an in-limited fill's output is clamped to the offer's own
11:58 UTC	xrpl-validator	`ca724f9`	fix(xrpl-ledger): F230 — the stream reaps an expired offer before the self-cross check, and steps on after a consumed fill
11:58 UTC	xrpl-validator	`5e46961`	fix(xrpl-ledger): F229 — NFTokenPage keys are the owner's account bytes, not a hash
14:08 UTC	xrpl-validator	`5d1a89d`	fix(xrpl-ledger): F240 — an XRP amount of more than sixteen digits is exact, in what rests and in the rate it rests at
14:08 UTC	xrpl-validator	`e37e205`	fix(xrpl-ledger): F241 — a check writer's shortfall is refused in preclaim, as tecPATH_PARTIAL

Findings fixed

- **fix(xrpl-ledger): F238 — an owner-directory insert sorts the whole page, not just the newcomer's slot** (`1274170`, xrpl-validator) rippled's dirAdd for an owner directory (preserveOrder == false — ApplyView.cpp directory::insertKey) sorts the last page's existing entries (`std::ranges::sort(indexes)`) and only then inserts the key at its lower bound. Pages written before SortedDirectories still hold their entries in insertion order until something is added to them, and the next insert then sorts every entry, not only the new one. Our dir_insert …
- **fix(xrpl-ledger): F239 — a bridged strand's rev pass reaps past the level it consumes whole** (`0ed3d96`, xrpl-validator) The direct walk has run rev_extent_reap per level since F114: rippled's rev pass steps a book by the strand's WANT, not by the taker's funds, so it consumes the head level whole and then steps — reaping every dead offer it passes until a live one on a worse level ends the iteration (*ofrQ != offer.quality(), BookStep.cpp:719). cross_bridged never did. Its only removals came from the F233 gate, which reaps where a …
- **fix(xrpl-ledger): F237 — a bridged round takes every leg-A offer at the tip's quality into one leg-B slice** (`ab1a5c4`, xrpl-validator) rippled's BookStep::forEachOffer consumes, in ONE pass, every live offer at the tip's quality (`execOffer` keeps going while `ofrQ` matches, up to the strand's limits), and the next step then prices their combined XRP once — for a pool on leg B, one slice for the whole level. Leg B of the bridged round has grouped its level this way since Finding 137; leg A still took one offer per round, so a level of several …
- **fix(xrpl-ledger): F236 — the direct walk rests an offer's in side as TakerGets less the gross fold divided once by the gateway rate** (`1ad42e1`, xrpl-validator) OfferCreate.cpp:490-517 rests what remains of a crossed offer's in side as `afterCross.in = takerGets - divideRound(result.actualAmountIn, gatewayXferRate, roundUp = true)`, where actualAmountIn is the sorted-multiset fold of the iterations' gross ins (Finding 226): one division of the total, rounded up. The bridged walk already rested it this way; the direct walk carried its running remainder — TakerGets less each …
- **fix(xrpl-ledger): F235 — NFTokenCreateOffer's preclaim is rippled's tokenOfferCreatePreclaim: an unfunded buyer is tecUNFUNDED_OFFER** (`c6751ea`, xrpl-validator) Our NFTokenCreateOffer preclaim checked the token's existence, the Expiration and the Destination/Owner disallow-incoming flags, and nothing else. rippled's nft::tokenOfferCreatePreclaim (NFTokenHelpers.cpp:811-882) adds, in this order: a transfer-fee NFT priced in an IOU needs its issuer alive, holding a line for that IOU unless it issues it, and not frozen on it (tecNO_ISSUER / tecNO_LINE / tecFROZEN); the account …
- **fix(xrpl-ledger): F234 — CheckCash cashes through the payment engine: SendMax buys what it can, DeliverMin is a floor, XRP shortfall is tecUNFUNDED_PAYMENT** (`d959488`, xrpl-validator) rippled cashes a check with the payment engine (CheckCash.cpp:331-577), not with a transfer of the requested amount: XRP xrpDeliver = DeliverMin ? max(DeliverMin, min(sendMax, srcLiquid)) : Amount with srcLiquid the writer's liquid XRP after its reserve for the check is released; a shortfall is tecUNFUNDED_PAYMENT.
- **fix(xrpl-ledger): F233 — bridge peeks reap dead offers only on the strands rippled flows this round** (`80eaadd`, xrpl-validator) rippled removes the expired and unfunded offers a stream steps past only inside a strand it FLOWS, and in an offer crossing a strand is flowed only when its quality upper bound — raw book tips (BookTip::step reads the first directory entry as it stands, dead or not) and pool spots — sits inside the limit (StrandFlow.h `qualityUpperBound < limitQuality => continue`). A fill taken on the direct strand says nothing …
- **fix(xrpl-ledger): F232 — a pool turn that ends the iteration leaves the tip in place; judge the beyond-strict sweep there, post-turn** (`f7ef2ee`, xrpl-validator) rippled's BookStep::forEachOffer consults the AMM once per iteration, at the iteration's first tip, and execOffer ends the iteration on the first offer whose quality differs from the anchor (`*ofrQ != offer.quality()`). So a pool turn whose slice quality sits off the level's own quality ends the

iteration BEFORE that level's offers are reached, and the NEXT iteration opens with its tip on the very same level — where …

- **fix(xrpl-ledger): F231 — an in-limited fill's output is clamped to the offer's own** (`e10f8d2`, xrpl-validator) Quality::ceilIn derives the output from the limited input at the offer's stored rate and then clamps it to the offer's own output (Quality.cpp ceilInImpl: "if (result.out > amount.out) result.out = amount.out"). A rate rounded to sixteen digits can otherwise hand out more than the offer holds, and our walk re-derived the maker's give from the clamped input without that bound.
- **fix(xrpl-ledger): F230 — the stream reaps an expired offer before the self-cross check, and steps on after a consumed fill** (`ca724f9`, xrpl-validator) OfferStream::step removes an EXPIRED offer before any self-cross or quality logic ever sees it (OfferStream.cpp:192, "Removing expired offer"), whoever owns it and wherever it sits against the limit. Our shared stepper tested a taker-owned offer against the limit first — right for a LIVE self-offer, which ends the pass — and so caught the taker's own expired offer and left it standing. And the bridged crossing's …
- **fix(xrpl-ledger): F229 — NFTokenPage keys are the owner's account bytes, not a hash** (`5e46961`, xrpl-validator) rippled's `keylet::nftokenPageMin(owner)` is the owner's twenty account bytes followed by twelve zero bytes and `nftokenPageMax` the same with 0xFF×12 (Indexes.cpp:418-431); a token's page is the base with the token's low 96 bits, and the owner's last page always sits at the max key. Our `nftoken_page_key` hashed `0x0050 || account` the way the indexHash keylets do, so finding 222's NFT-holder check on AccountDelete …
- **fix(xrpl-ledger): F240 — an XRP amount of more than sixteen digits is exact, in what rests and in the rate it rests at** (`5d1a89d`, xrpl-validator) A native STAmount carries drops as an exact integer up to 1e17. It is not held to the sixteen-digit mantissa an IOU is canonicalized into, and two places in the resting path assumed otherwise.
- **fix(xrpl-ledger): F241 — a check writer's shortfall is refused in preclaim, as tecPATH_PARTIAL** (`e37e205`, xrpl-validator) rippled checks the writer's balance before the payment engine ever runs (CheckCash.cpp:169-192): availableFunds = accountFunds(check.Account, value, ZeroIfFrozen, ZeroIfUnauthorized) if (value.native() && !check[sfSponsor]) availableFunds += fees().increment if (value > availableFunds) return tecPATH_PARTIAL
- **fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties** (`3fefb0d`, xrpl-validator) rippled unlinks a DepositPreauth with keepRoot FALSE: view.dirRemove(keylet::ownerDir(account), page, preauthIndex, false) DepositPreauth.cpp:283

Independence program

No independence record for UTC day 2026-09-09; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

Open items

- Receipt R-03 (#106,864,621): open, under investigation.
- Receipt R-06 (#106,869,040): open, under investigation.
- Receipt R-07 (#106,869,060): open, under investigation.
- Receipt R-09 (#106,869,323): open, under investigation.
- Receipt R-13 (#106,874,385): open, under investigation.
- Receipt R-14 (#106,874,386): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Confidential Vault Share Balances** (lending) Add an lsfVaultConfidential flag to the Vault ledger object (SingleAssetVault); when set, VaultDeposit/VaultWithdraw store each depositor's position a
- **BLS Light-Client Verified Bridge Headers** (interop) Add an owner-reserved ledger object ltXCHAIN_LIGHT_CLIENT (keylet derived from the Bridge object plus door account) holding a counterparty chain's tru
- **Credential-Bound Session Key Revocation** (accounts) Extend g11's SessionKey ledger object with optional SFields sfCredentialType and sfCredentialIssuer, settable only at creation via SessionKeySet. When

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-09 05:24	09-09 11:58	xrpl-validator	`1274170`	fix(xrpl-ledger): F238 — an owner-directory insert sorts the whole page, not just the newcomer's slot

Committed	Merged into main	Repo	Commit	Subject
09-09 11:52	09-09 11:58	xrpl-validator	`0ed3d96`	fix(xrpl-ledger): F239 — a bridged strand's rev pass reaps past the level it consumes whole
09-09 02:24	09-09 11:58	xrpl-validator	`ab1a5c4`	fix(xrpl-ledger): F237 — a bridged round takes every leg-A offer at the tip's quality into one leg-B slice
09-08 22:54	09-09 11:58	xrpl-validator	`1ad42e1`	fix(xrpl-ledger): F236 — the direct walk rests an offer's in side as TakerGets less the gross fold divided once by the gateway rate
09-08 18:16	09-09 11:58	xrpl-validator	`c6751ea`	fix(xrpl-ledger): F235 — NFTokenCreateOffer's preclaim is rippled's tokenOfferCreatePreclaim: an unfunded buyer is tecUNFUNDED_OFFER
09-08 17:04	09-09 11:58	xrpl-validator	`d959488`	fix(xrpl-ledger): F234 — CheckCash cashes through the payment engine: SendMax buys what it can, DeliverMin is a floor, XRP shortfall is tecUNFUNDED_PAYMENT
09-08 16:47	09-09 11:58	xrpl-validator	`80eaadd`	fix(xrpl-ledger): F233 — bridge peeks reap dead offers only on the strands rippled flows this round
09-08 16:26	09-09 11:58	xrpl-validator	`f7ef2ee`	fix(xrpl-ledger): F232 — a pool turn that ends the iteration leaves the tip in place; judge the beyond-strict sweep there, post-turn
09-08 15:08	09-09 11:58	xrpl-validator	`e10f8d2`	fix(xrpl-ledger): F231 — an in-limited fill's output is clamped to the offer's own
09-08 12:45	09-09 11:58	xrpl-validator	`ca724f9`	fix(xrpl-ledger): F230 — the stream reaps an expired offer before the self-cross check, and steps on after a consumed fill
09-08 12:32	09-09 11:58	xrpl-validator	`5e46961`	fix(xrpl-ledger): F229 — NFTokenPage keys are the owner's account bytes, not a hash
09-09 13:22	09-09 14:08	xrpl-validator	`5d1a89d`	fix(xrpl-ledger): F240 — an XRP amount of more than sixteen digits is exact, in what rests and in the rate it rests at
09-09 14:08	09-09 14:08	xrpl-validator	`e37e205`	fix(xrpl-ledger): F241 — a check writer's shortfall is refused in preclaim, as tecPATH_PARTIAL
09-09 15:03	—	xrpl-validator	`3fefb0d`	fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties

Times in UTC. A commit appears on the day it was committed to main or merged into it.